Morning Brief

Ludovici · Thursday, June 11, 2026

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By Stephanie Ludovici

A personal learning and awareness brief: AI governance, cybersecurity, project status, five new project ideas, and four training items. Built to keep me current and sharp.

Email

Gmail is connected (read-only); Google Calendar is not yet connected — the calendar section will appear here once it is.

Mostly promotional traffic overnight. The few items worth a glance:

- **Neibauer Dental (Harrison Crossing)** — two messages: a reminder for yesterday's 4:00 PM appointment (June 10) and a confirmation for a new appointment **Monday, July 6 at 3:00 PM**. Worth adding to your calendar once it's connected.
- **ADT** — routine panel activity (disarm by Julie Rosenburg at 5:43 AM; front-door sensor-left-open restoral last evening) plus an ADT "review who can access your system" nudge — a reasonable prompt to prune stale logins.
- **Sent item to mail.mil** (you → stephanie.e.ludovici.civ) re: rotation opportunities, where you flagged an approved reasonable accommodation. Logged here for your own tracking; no action needed from this brief.

- **Professional/learning:** ISC2 Security Congress 2026 agenda is live (Oct 24–28); Elastic and Elastic/Towards Data Science newsletters both leading with RAG-hallucination themes — adjacent to the v3.0 groundedness overlay you're tracking.

Triage take: nothing urgent or security-sensitive. The dental appointment is the only calendar-worthy item.

AI governance & policy

- **OMB issues M-26-04, "Increasing Public Trust in AI Through Unbiased AI Principles."** A new federal memo framing agency AI use around bias/trust principles — directly relevant to how a tiered IC framework articulates fairness and accountability obligations. Worth reading against your evaluation-dimensions section (06). [OMB AI policy overview (OneTrust summary)]
- **NIST stands up a dedicated autonomous-AI-agent standards initiative (launched Feb 2026).** NIST is now developing standards for systems that take real-world actions without continuous human oversight — the exact gap your v3.0 agentic overlay flags as single-sourced to OWASP. A second authoritative anchor to triangulate against. [NIST AI / governance landscape (eccouncil)]
- **EU publishes a Code of Practice on marking and labeling AI-generated content (June 10, 2026).** Provenance-marking guidance lands just as your v2.1 pulled AI-output provenance marking forward (§7.1). Useful external comparator even though it's EU, not IC. [EU AI Act regulatory framework]
- **EU AI Act becomes fully enforceable August 2, 2026; high-risk-area rules apply December 2, 2027.** The near-term enforcement date sharpens the global baseline for high-risk classification schemes — a benchmark for tier definitions. [EU AI Act developments tracker]
- **Treasury's financial-services framework maps NIST AI RMF into 230 control objectives.** A concrete worked example of translating RMF principles into auditable controls — a template pattern for crosswalking RMF to your required-artifacts list. [AI governance & compliance 2026 guide]

Cybersecurity & threats

- **CISA adds three KEVs (June 9, 2026):** CVE-2026-7473 (Arista EOS incomplete comparison), CVE-2026-11645 (Chromium V8 out-of-bounds read/write), CVE-2026-20245 (Cisco Catalyst SD-WAN Manager improper output encoding). The Chromium and Cisco SD-WAN items have broad federal-estate exposure; check patch posture. CISA KEV addition, June 9
- **BOD 26-04 — "Prioritizing Security Updates Based on Risk."** CISA's new directive evolves the KEV model: prioritize high-risk vulnerabilities for timely remediation, defer low-risk ones. This changes how FCEB agencies sequence patching versus the flat BOD 22-01 due-date model. BOD 26-04
- **CISA launches a new reporting form for Known Exploited Vulnerabilities.** A process change for how exploited-vuln evidence is submitted — minor operationally, but relevant if you reference KEV intake in any reachability/monitoring spec. CISA KEV reporting form
- **Standing KEV catalog context.** BOD 22-01 still requires FCEB agencies to remediate cataloged vulnerabilities by their due dates; BOD 26-04 layers risk-based prioritization on top rather than replacing it. CISA KEV catalog

Threat-model tie-in: NetMon's loopback-only, no-external-network design keeps it off the SD-WAN/ browser exposure path entirely — a good reminder that its constraint posture is a feature, not a limitation.

Project status

Three most recently touched projects under `Projects\`:

- **NetMon-Server** (v0.40.2, June 9) — Active. Last work fixed chart-panel help "?" tooltips that were clipped by `content-visibility` paint containment; routed through the shared body-anchored `positionSectionTip`. Verified via PS parse + `node --check`. **Next:** add a regression check (or screenshot diff) for tooltip positioning so the anchor fix doesn't silently regress on the next `content-visibility` change.
- **Radio AI — SEL** (v1.0.1, June 8; backlog updated June 9) — Active sandbox/dev copy, simulator-first. BACKLOG is the single source of truth (epics `SEL-<epic><n>`), including a continuous security workstream and an AI-agent roadmap. **Next:** pick the

top `SEL-A` story and cut a simulator test-harness slice before adding the spectrum scope, so new features land against a test baseline.

- **AI Framework** (roadmap/README current) — Stable; v3.0 work is deliberately deferred. Today's news gives you two concrete anchors for the forward items: NIST's new **autonomous-AI-agent standards initiative** directly supports the §2 agentic overlay (currently single-sourced to OWASP), and the EU **content-marking Code of Practice** is an external comparator for the v2.1 provenance-marking requirement (§7.1). **Next:** open a v3.0 evidence-verification note that records the NIST agent-standards initiative as the second source the roadmap says it needs before drafting any agentic control.

Five new project ideas

Full write-ups (with method, objective, and duration) saved to `project-ideas\2026-06-11-ideas.md`. Summary:

1. **Geospatial foundation-model fine-tuning for wildfire burn-severity mapping** — *Remote sensing / ecology; Python.* **Problem:** burn-severity maps are slow and label-hungry. **Objective:** measure how much labeled data a Prithvi-EO backbone actually saves vs a scratch U-Net. **Method:** fine-tune frozen-then-unfrozen geospatial ViT with a segmentation head on Sentinel-2/HLS; MTBS dNBR as silver labels; learning curves at 1/10/100% labels. **Duration:** 3–4 weeks.
2. **Weakly supervised anomaly detection for new-physics searches (LHC Olympics)** — *HEP; Python + R.* **Problem:** unpredicted signals are missed by hypothesis-specific searches. **Objective:** compare autoencoder vs CWoLa weak-supervision sensitivity without signal labels. **Method:** jet-feature engineering; reconstruction-error and sideband-vs-signal-region classifiers; SIC curves, significance bootstrapped in R. **Duration:** 4–5 weeks.
3. **Species distribution modeling under climate scenarios with spatial cross-validation** — *Ecology; R primary.* **Problem:** random splits leak spatial autocorrelation and inflate SDM accuracy. **Objective:** quantify the random-vs-spatial-block CV gap, then project to CMIP6 futures. **Method:** GLM/RF/boosted trees in tidymodels; `spatialsample` block CV; terra projection to SSP scenarios. **Duration:** 2–3 weeks.

4. **Symbolic regression to rediscover governing equations** — *Physics / dynamical systems; Julia (SymbolicRegression.jl) + PySR.* **Problem:** black-box fits give no interpretable law. **Objective:** map when symbolic regression recovers true equations vs noise and sample size. **Method:** simulate trajectories with controlled noise; Pareto accuracy-vs-complexity search; exact-recovery rate over noise × n grids; validate on real materials data. **Duration:** 2–3 weeks.
5. **Protein variant-effect prediction with frozen protein-language-model embeddings** — *Genomics; Python + R.* **Problem:** deleteriousness prediction is heavy, and calibration is under-examined. **Objective:** benchmark frozen ESM-2 embeddings + light heads vs the zero-shot log-likelihood baseline on ProteinGym, focusing on calibration. **Method:** per-variant embeddings; LR/MLP heads with protein-held-out splits; Spearman/AUROC plus reliability curves and ECE in R. **Duration:** 3–4 weeks.

Four training items

1. Python — context managers and `contextlib`

A context manager guarantees setup/teardown around a block via the `with` statement, so resources (files, locks, DB connections, temporary state) are released even when the block raises. You've used `with open(...)`, but the more useful skill is writing your own. The lightest way is the `@contextmanager` decorator: everything before `yield` is setup, everything after is teardown, and wrapping the `yield` in `try/finally` makes cleanup exception-safe.

```
import time
from contextlib import contextmanager


@contextmanager
def timed(label):
    # Setup runs when the with-block is entered.
    start = time.perf_counter()
    try:
        # The value after yield is bound to the "as" targ
        yield
    finally:
        # Teardown runs on normal exit AND on exception,
        elapsed = time.perf_counter() - start
        print(f"{label}: {elapsed:.4f}s")
```

```
with timed("vector sum"):
    total = sum(range(10_000_000))  # Work whose duration
```

Why it matters: it moves cleanup logic next to acquisition and out of the caller's hands — fewer leaked file handles and unreleased locks. **Common pitfall:** putting the cleanup *after* the `yield` without a `try/finally`. If the body raises, the post-`yield` lines never run and the resource leaks. Always wrap the `yield`. Reference: [Python docs — `contextlib`]

2. R — reshaping with `tidyr` `pivot_longer()` / `pivot_wider()`
Tidy data wants one observation per row and one variable per column, but real tables arrive "wide" (a column per month) or too "long." `pivot_longer()` and `pivot_wider()` are the modern, idiomatic replacements for the old `gather/spread` and for `reshape2`. Most analysis bugs in R trace back to data that's the wrong shape for `ggplot2` or `dplyr`.

```
library(tidyr)
library(dplyr)

# Wide: one row per site, a column per quarter.
wide <- tibble::tibble(
  site = c("A", "B"),
  q1   = c(10, 7),
  q2   = c(12, 9)
)

# Pivot to long form so quarter becomes a single variable
long <- wide |>
  pivot_longer(cols = q1:q2, names_to = "quarter", values

# Aggregate in long form, then pivot back to a compact re
report <- long |>
  group_by(quarter) |>
  summarise(mean_reading = mean(reading), .groups = "drop
  pivot_wider(names_from = quarter, values_from = mean_re
```

Why it matters: `dplyr` verbs and `ggplot2` aesthetics assume long, tidy data; reshaping first removes whole classes of grouping/faceting errors. **Pitfall:** selecting the wrong columns in `cols =` and accidentally pivoting an ID column into the value set — name the columns explicitly (`q1:q2`) or use selectors like `starts_with("q")`. Reference: [tidyr — pivoting vignette]

3. Data-centric — idempotent pipelines and the upsert/MERGE pattern

A pipeline is **idempotent** when running it twice with the same input produces the same result as running it once. Without idempotency, a retried or backfilled job double-counts rows and corrupts downstream metrics. The core tool is the **upsert** (insert-or-update on a natural/business key), expressed in SQL as `MERGE` or `INSERT ... ON CONFLICT`. The job becomes "make the target match the source for these keys," not "append whatever I computed."

```
-- Upsert on the business key so re-running the job is sa
MERGE INTO dim_device AS tgt
USING staging_device AS src
  ON tgt.device_id = src.device_id          -- Natural ke
WHEN MATCHED THEN UPDATE SET
  tgt.hostname   = src.hostname,
  tgt.updated_at = src.loaded_at
WHEN NOT MATCHED THEN INSERT (device_id, hostname, update
  VALUES (src.device_id, src.hostname, src.loaded_at);
```

Why it matters: orchestrators (Airflow, cron, manual reruns) *will* re-execute a task; an idempotent upsert makes retries and backfills harmless instead of dangerous. **Design notes:** pick a stable business key (a surrogate auto-increment defeats the purpose), and keep the staging→target load deterministic so a replay reconstructs identical state. Reference: dbt — idempotency & incremental models

4. AI method of the day — Random Fourier Features (RFF)

Kernel methods like RBF kernel ridge regression are accurate but scale as $O(n^2)$ memory / $O(n^3)$ time because they form the full kernel matrix. **Random Fourier Features** sidestep that: by Bochner's theorem, a shift-invariant kernel is the Fourier transform of a probability measure, so you can draw random frequencies $w \sim N(0, \gamma^2 I)$ and phases $b \sim U(0, 2\pi)$ and build an explicit map $z(x) = \sqrt{(2/D)} \cdot \cos(w^\top x + b)$ whose inner products approximate the kernel: $E[z(x) \cdot z(x')] \approx k(x, x')$. You then fit a *linear* model in that D-dimensional space at O(nD) cost, recovering kernel-quality fits at a fraction of the price. **When to use it:** you want RBF-kernel behavior but n is too large for an $n \times n$ Gram matrix, or you need an explicit feature vector to feed a linear/streaming learner. **Intuition for D:** more random features means a tighter kernel approximation — held-out error falls then plateaus toward the exact kernel.

Demonstrated end-to-end in today's companion notebook — `notebooks\2026-06-11-random-fourier-features.ipynb` — with explicit **Modeling & Simulation** (noisy nonlinear signal, train/test split) and **Test & Evaluation** (held-out RMSE and R^2, a residual-based 90% prediction interval with empirical coverage, and a D-sweep confirming the accuracy plateau). Pure NumPy + matplotlib; outputs cleared. Reference: [Rahimi & Recht, "Random Features for Large-Scale Kernel Machines," NeurIPS 2007]

Novelty ledgers updated: five idea titles appended to `project-ideas\_ledger.md`, four training topics appended to `training-ledger.md`.

Attached: brief (.pdf) + companion notebook (.ipynb) + project briefs (.docx)
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